

Appendix of Embodied Robot Manipulation in the Era of Foundation Models: Planning and Learning Perspectives

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Abstract—In this supplementary document, we provide additional discussions, clarifications, and comparisons:

- How foundation models benefit robotic manipulation compared with conventional methods (see Section A).
- A summary of major manipulation task types and the corresponding representative methods (see Section B).
- A clarification of generalization in foundation-model-based robotic manipulation, together with representative methods that address this challenge (see Section C).
- A quantitative cross-benchmark comparison of representative low-level action models and a controlled empirical analysis of key VLA design choices (see Section D).

APPENDIX

A. Comparison Between Foundation Model-Based and Conventional Methods

Compared with conventional manipulation pipelines, foundation models broaden robotic manipulation from a largely

task-specific control problem to one that tightly couples semantic understanding, spatial grounding, and action generation. Traditional methods often rely on carefully engineered task representations, manually designed rewards or constraints, and fixed object or task settings, which limits their flexibility in open-ended environments. By contrast, foundation models introduce large-scale semantic and multimodal priors that enable robots to interpret free-form language instructions, reason over unseen objects and scenes, infer task-relevant affordances or subgoals, and support more transferable policy learning. Their contribution to manipulation is therefore not merely improved perception, but a new interface between high-level intent understanding and downstream motion generation.

Problem Formulation. Conventional manipulation methods are typically formulated around well-specified control objectives, such as trajectory optimization, feedback control, reward maximization, or expert action imitation. In contrast, foundation model-based methods often reformulate manipulation as a joint problem of instruction understanding, semantic grounding, intermediate reasoning, and action generation. This shift is particularly important in open-world settings, where the robot must determine not only how to execute a motion, but also what to do, where to act, and in what order.

Capability Differences. Foundation model-based methods differ from conventional approaches in several important aspects. First, they enable more flexible task specification through natural language, reducing the reliance on manually engineered symbolic interfaces or task-specific reward design. Second, they improve semantic generalization across unseen objects, instructions, and scenes by leveraging large-scale pretraining. Third, they provide stronger support for long-horizon and compositional manipulation, since they can produce intermediate artifacts such as task plans, affordance priors, subgoals, code, or structured scene descriptions that guide downstream execution. By contrast, conventional methods are often more specialized and may require explicit redesign when the task distribution changes.

Relationship to Conventional Execution Modules. Foundation models do not completely replace conventional motion generation and control. In most robotic manipulation systems, they primarily contribute high-level reasoning, semantic abstraction, or structured intermediate guidance, while low-level trajectory generation and control are still handled by motion planners, trajectory optimization methods, feedback controllers, or learned action policies. This is because precise execution remains constrained by embodiment, dynamics, safety, latency, and closed-loop stability, all of which are areas

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TABLE I
COMPARISON BETWEEN NON-LEARNING AND LEARNING-BASED METHODS FOR GRASPING AND MANIPULATION. HERE, GRASPING SPECIFICALLY
REFERS TO GRASP DETECTION AND GENERATION.

Mani. Type	Control Type	Pose Generation	Trajectory Generation	Generalization	Interpretability & Stability
Grasping	Non-learning	Analytical: geometric rules, force-closure analysis Learned from data	IK + motion planning	Low	High
	Learning		IK + motion planning	High	Low
Manipulation	Non-learning	Task-specific or predefined goal pose	IK + motion planning	Low	High
	Learning	Learned implicitly via RL or IL	Learned policies via RL or IL	High	Low

where conventional control methods remain essential.

Limitations and Failure Modes. Foundation model-based manipulation methods also introduce new challenges. Their semantic predictions may not always correspond to physically feasible actions; inferred affordances or subgoals may be plausible at the language level but geometrically invalid; and long-horizon reasoning still often suffers from error accumulation. In addition, large models often incur higher computational cost and latency, which may conflict with strict real-time manipulation. These limitations suggest that foundation models and conventional methods are best viewed not as competing paradigms, but as complementary components in a hierarchical manipulation stack.

B. Manipulation Tasks

Early grasping methods primarily focused on identifying stable grasp configurations through geometric analysis, force closure conditions, or task-specific heuristics. Candidate grasp poses were analytically generated from object geometry, contact normals, or predefined grasp templates, and then executed using inverse kinematics (IK) and motion planning. Building on such established grasps, early approaches to dexterous, deformable, mobile, quadrupedal, and humanoid manipulation typically assumed a known target pose or a task-specific goal. The emphasis was on optimizing motion trajectories or control commands to achieve these goals under physical and kinematic constraints, rather than learning end-to-end action generation from raw sensory input as in modern RL or IL methods. The comparison is summarized in Table I.

1) *Grasping*: In the narrow sense considered in this work, grasping specifically refers to the tasks of grasp detection and grasp generation. These tasks involve identifying feasible grasp configurations from sensor inputs such as images or point clouds, allowing robotic end-effectors to securely pick up objects. The primary focus is on predicting the position and orientation of the gripper to ensure stable and reliable grasps, even in the presence of diverse object shapes, varying poses, and cluttered environments.

Non-Learning-based Grasp. Early methods generate grasp poses by explicitly analyzing object geometry [1], performing contact-driven force analysis [2], or applying task-specific rules [3], in combination with the gripper model. However, due to their limited generalization ability in handling complex shapes, occlusions, and unseen objects, these approaches have gradually been replaced by data-driven, learning-based methods, as discussed below. The taxonomy of learning-based grasping approaches is illustrated in Figure 1.

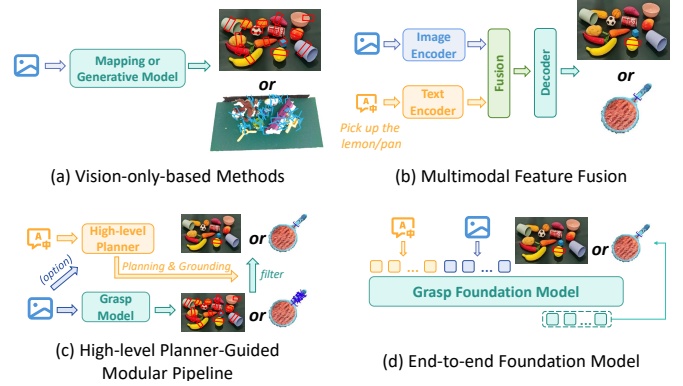


Fig. 1. Comparison of different grasping methods. (a) Vision-only approaches, which map visual inputs to grasp poses using CNNs or spatial transformers, or generate poses via VAEs or diffusion models. With the introduction of language, three main categories emerge: (b) fusion of language and visual features through mechanisms such as cross-attention; (c) generation of multiple grasp candidates using pretrained grasp models, followed by selection using high-level reasoning modules, such as LLMs, VLMs, or 3D scene representations; and (d) end-to-end fine-tuning of grasp foundation models on large-scale grasping datasets. Figure adapted from [13]–[15].

Rectangle-based Grasp. Grasping rectangles were first introduced by Jiang et al. [4]. While they are visually similar to bounding boxes, their representational semantics are fundamentally different. A grasping rectangle is defined as a 5-dimensional representation. Subsequent methods have progressively incorporated deep learning techniques, ranging from basic convolutional neural networks (CNNs) [5]–[7] to more advanced architectures such as ResNet [8], GR-ConvNet [9] and Transformer [10], and further to CLIP-based models that integrate textual information through feature fusion methods [11]. More recently, diffusion models have also been employed [12]. Over time, the models have evolved from simple to complex, and the modalities have shifted from unimodal to increasingly multimodal.

6-DoF Grasp. Two-dimensional rectangle-based representations, which typically assume parallel-jaw grippers executing top-down grasps, are limited to 2 or 3 degrees of freedom (DoF). This restricts the gripper’s orientation and reduces applicability in unstructured or complex environments. To address these limitations, researchers have proposed 6-DoF grasp representations [16] that define a grasp as a complete 6-dimensional pose in 3D space. This formulation enables the robot to grasp objects from arbitrary orientations, significantly improving flexibility and robustness. The 6-DoF representation is especially beneficial in cluttered scenes, non-planar object poses, and scenarios involving complex geometries.

Apart from a few early 2D-based approaches [17], the majority of recent methods adopt 3D representations. In particular, methods based on point cloud inputs have explored a wide range of architectures, ranging from mapping convolutional networks [18] and transformer-based models [19], to generative autoencoders [16], variational autoencoders [20], [21], and UNet-style frameworks [22]. To improve efficiency, various methods have been proposed, including applying instance segmentation or heatmap to focus on task-relevant manipulation regions [23], [24], reducing the grasp search space by representing grasp poses with contact points on the object surface [25], incorporating graph neural networks for geometric reasoning on point cloud data [26], and adopting an economic supervision paradigm that selects key annotations and leverages focal representation to reduce training cost and improve performance [27]. Some works further address structural distortions commonly found in real-world point cloud data by introducing completion or denoising modules to convert the input into a clean and consistent style [28], [29]. Several methods also address the SE(3)-equivariance problem [30], [31] by modeling grasp generation as a continuous normalizing flow over SE(3) with equivariant vector fields, or by predicting per-point grasp quality over the orientation sphere using spherical harmonic basis functions. Meanwhile, several methods also leverage 3D representations such as SDF [32]–[34], NeRF [35], [36], and 3DGS [37]–[39] to extract geometric features or to sample grasp points for downstream prediction.

Language-driven Grasp. In recent years, language-driven grasping has gained increasing attention, aiming to achieve object-specific and instruction-guided manipulation. Existing approaches can be broadly categorized into three groups. The first adopts multimodal feature fusion [40], [41], where textual and visual modalities are jointly encoded, often via cross-attention mechanisms. The second leverages existing grasp models to generate large numbers of grasp candidates, followed by ranking or scoring with LLMs or VLMs to select the most confident grasps. For instance, LLMs can generate task-specific descriptions [42], while VLMs are used for visual grounding to compute grasp confidence scores [15], [43]. Representative methods include VL-Grasp, which employs VLMs to attend to the target object and generate grasps [44]; OWG, which leverages VLM-based semantic priors for planning under occlusion [45]; and Reasoning-Grasping, which integrates visual-linguistic inference for improved object-level understanding [46]. Other efforts adapt MLLMs for environment-aware error correction [47], or learn object-centric attributes to enable rapid grasp adaptation across tasks [48]. Affordance-driven approaches also ground grasp generation in language and vision cues [49]–[51]. The third line of work directly fine-tunes MLLMs on large-scale grasping datasets to build grasp foundation models [13], [52].

Challenges. First, the aforementioned grasp detection and generation methods were originally designed for 2-DoF parallel-jaw grippers. However, with the increasing use of dexterous hands, research has shifted toward dexterous grasping, which requires more complex annotations such as hand joint angles and contact force maps. To handle the high dimensional-

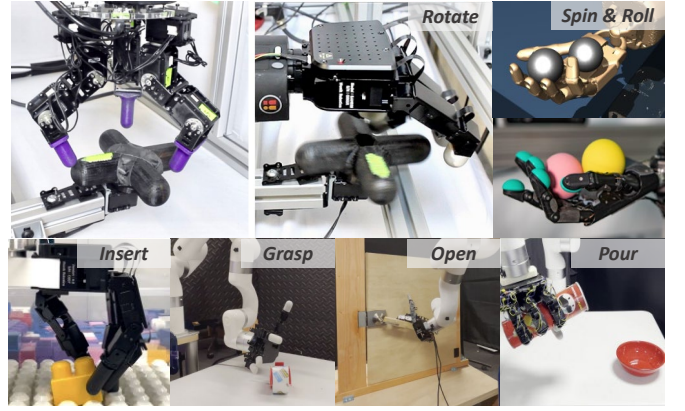


Fig. 2. Tasks in dexterous manipulation, figure adapted from [64], [66], [103]–[105].

ity of this task, various generative approaches—including diffusion-based models—have been proposed [53]–[55]. Some approaches represent grasp configurations via contact points or maps [56], or leverage interaction-based representations like $D(R, O)$ [57] to infer grasps from geometric relationships. Second, traditional grasping strategies typically rely on a single end-effector (e.g., suction or parallel grippers), which limits adaptability to diverse object geometries. To address this, several works have explored bimanual grasping using dual-arm or dual-gripper configurations [58], [59]. Finally, grasping transparent objects remains a significant challenge, as depth sensors often fail to detect or localize such materials accurately. Recent efforts address this limitation by incorporating alternative modalities, such as LiDAR [60] or 3D reconstruction [61]–[63], to infer the geometry of transparent or reflective objects.

2) Dexterous Manipulation: Dexterous manipulation refers to the capability of robotic systems equipped with multi-fingered or anthropomorphic hands to achieve precise and coordinated object control through complex contact interactions. It involves in-hand reorientation, fine force modulation, and multi-point contact, enabling actions such as twisting, grasping, and rotating, as illustrated in Figure 2. This capability is critical for tasks requiring high precision and adaptability, such as tool use, assembly, and manipulation of small or irregular objects. Human hand models typically include 20–25 DoF, with each finger modeled by 4 DoF, the thumb by 4–5 DoF, and additional DoF from the palm and wrist for enhanced realism [102].

Non-Learning-based Methods. Early work on dexterous manipulation predominantly employed non-learning approaches, including optimization- and control-based techniques such as heuristic search and constrained optimization [106]. These methods typically assume access to known dynamics and object models, and focus on trajectory planning under physical constraints.

Learning-based Methods. With the advent of RL, both model-based and model-free paradigms have been applied to dexterous manipulation. Model-based methods such as PDDM [64] exploit learned dynamics for efficient planning and control, while model-free approaches directly optimize

TABLE II
REPRESENTATIVE METHODS ACROSS MANIPULATION TYPES AND LEARNING PARADIGMS.

Mani. Type	RL	IL	RL+IL	VLA
Dexterous	PDDM [64], [65], [66]	DexHandDiff [67], CordViP [68]	REBOOT [69], ViViDex [70]	OFA [71], LBM [72]
Soft Robotics	[73], [74]	Soft DAgger [75], KineSoft [76]	SS-ILKC [77]	–
Deformable Object	[78]	DeformerNet [79], DexDeform [80]	DMfD [81]	–
Mobile	[82], MoMa [83]	MOMA-Force [84], Skill Transformer [85]	[86]	MoManipVLA [87]
Quadrupedal	VBC [88], GAMMA [89]	Human2LocoMan [90]	[91], WildLMa [92]	QUAR-VLA [93], GeRM [94]
Humanoid	[95], FLAM [96]	OmniH2O [97], iDP3 [98]	[99]	GR00T N1 [100], Humanoid-VLA [101]

policies from interaction [65], sometimes enhanced with few-shot imitation to accelerate learning on real hands [66]. More recently, human-in-the-loop frameworks such as HIL-SERL [107] combine demonstrations, online corrections, and reinforcement learning to achieve sample-efficient training of precise and dexterous skills directly on real robots. IL has gained attention for its data efficiency, exemplified by DexMV [108], which employs kinematic retargeting from human videos to robot hands. Extensions incorporate auxiliary tasks such as contact prediction [67], [68] and inverse dynamics modeling [109], enabling better generalization and action consistency. Hybrid IL–RL approaches aim to combine these strengths, typically using IL for policy initialization followed by RL refinement [110], [111]. Integration can vary: DexMV applies IL first and fine-tunes with RL, whereas ViViDex [70] performs RL in a privileged state space before distilling policies via IL.

Recent advances in VLA models further expand generalization. DexGraspVLA [112] integrates semantic reasoning with diffusion-based policies for grasping in cluttered scenes, while OFA [71], LBM [72], and Being-H0 [113] leverage multimodal prompts and human videos to generate dexterous motions and follow language instructions.

Human guidance provides another supervision source: Chen et al. [114] use object and wrist trajectories from videos to guide RL, and Mandi et al. [115] introduce functional retargeting to transition from demonstrations to autonomous control. Affordance reasoning further supports grasp-specific strategies [116], [117], guiding object selection under semantic constraints. Finally, recent works emphasize robustness and autonomy. Task decomposition reduces complexity by structuring subtasks [118], while recovery mechanisms such as REBOOT [69] introduce IL-trained reset policies to handle failures in long-horizon settings.

Challenges. Beyond high-dimensional control and contact dynamics, dexterous manipulation faces broader challenges. First, many real-world tasks are long-horizon and compositional, requiring sequential execution of fine-grained skills. Chen et al. [104] address this by decomposing tasks into discrete skills trained with RL and linking them via a high-level policy. Second, sim-to-real transfer remains a bottleneck due to discrepancies in perception, dynamics, and actuation. CyberDemo [105] mitigates this through data augmentation, improving robustness under domain shift. Third, accurate perception is difficult in cluttered or partially observed scenes, where occlusion hampers object tracking. DexPoint [103] leverages point cloud completion to recover missing geometry and enhance spatial awareness.

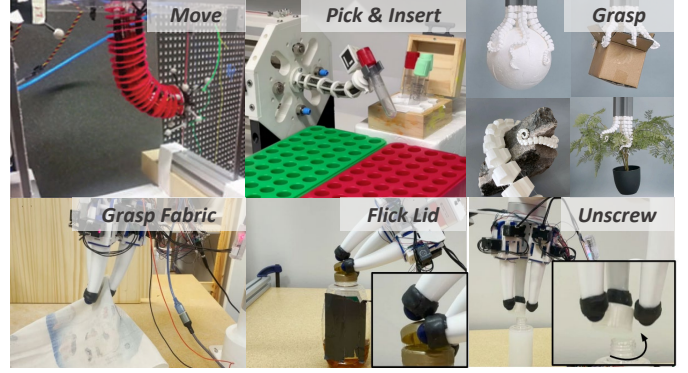


Fig. 3. Tasks in soft robotic manipulation, figure adapted from [76], [124], [125].

3) *Soft Robotic Manipulation:* Soft manipulators, built with compliant materials or structures, are well-suited for human–robot collaboration, operation in uncertain environments, and safe, adaptive grasping, as illustrated in Figure 3. They overcome the limitations of rigid manipulators when handling delicate or deformable objects [119]. To this end, a wide range of designs have been developed to support diverse applications [120]–[124].

Non-Learning-based Methods. For soft manipulator control, analytical kinematic models based on the constant-curvature assumption map shape parameters to end-effector poses via virtual rigid-link chains and Denavit–Hartenberg transformations [126]. Similarly, three-dimensional continuum manipulators can be approximated as rigid-jointed robots, with computed-torque control achieving closed-loop dynamics [127]. Other approaches employ model predictive control with Koopman operators [128], [129], or rely on accurate Lagrangian models combined with adaptive dynamic sliding mode control to enhance robustness [130]. Hybrid schemes also integrate learning into non-learning frameworks: forward dynamics can be approximated with machine learning and combined with trajectory optimization for open-loop control [131], while feedback-driven strategies exploit learned models to stabilize and restore system states [121].

Learning-based Methods. RL and IL have been widely applied to soft manipulator control. Model-free RL avoids explicit physical modeling by training policies in simulation for closed-loop endpoint control [125]. Forward dynamics learned with recurrent networks can be integrated into trajectory optimization to generate samples and train predictive controllers [73], while LSTM-based dynamics models enable feedback policy learning [132]. Domain randomization com-

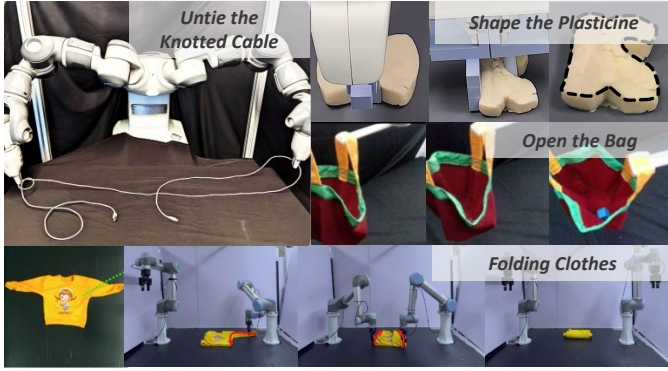


Fig. 4. Tasks in deformable object manipulation, figure adapted from [134]–[137].

bined with incremental offline training has improved task-space accuracy and adaptability [74], and wavelet-based dynamics approximations paired with LSTM and TD3 controllers further enhance generalization under variability [133]. IL approaches include Soft DAgger [75], which employs dynamic behavior mapping for online expert-like action generation, and KineSoft [76], which integrates kinesthetic teaching with diffusion-based policy learning. Hybrid methods also emerge, such as SS-ILKC [77], which combines multi-objective RL with adversarial IL to learn goal-directed control strategies in sensor space, supported by sim-to-real pre-calibration for zero-shot transfer.

Challenges. Soft-hand-based manipulation faces several challenges, including modeling highly nonlinear and underactuated dynamics, achieving precise force and pose control with deformable or fragile objects, ensuring robustness under environmental uncertainty and sensor noise, and lowering the cost and complexity of data collection and teleoperation for policy training. To improve sample efficiency, Soft DAgger [75] enables online imitation learning from limited demonstrations through dynamic behavior mapping. To reduce hardware complexity in teleoperation, Liu et al. [123] propose a flexible bimodal sensory interface that combines vision-based perception with wearable sensors, enabling intuitive and low-cost control without bulky equipment.

4) *Deformable Object Manipulation:* Deformable object manipulation (DOM) requires robots to perceive and control non-rigid objects whose shapes vary under applied forces. Unlike rigid-body manipulation, it demands reasoning over high-dimensional, continuous state spaces, coping with uncertain deformation dynamics, and interpreting subtle visual or tactile cues. As illustrated in Figure 4, tasks such as cloth folding, rope and cable tying, and food handling involve diverse deformations—including tension, compression, and bending—making DOM a complex yet crucial challenge in robotic manipulation [138]–[140].

Non-Learning-based Methods. Traditional approaches to DOM, such as path planning [141] and model-based control [142], often rely on simplified physical models or analytical solutions. However, these methods typically struggle with generalization and real-time adaptability in complex environments, leading to a shift toward data-driven techniques

such as RL, IL, and hybrid learning-control frameworks.

Learning-based Methods. Matas et al. [78] propose a deep RL method based on a modified DDPG algorithm that learns from visual and robot state inputs and achieves zero-shot sim-to-real transfer via domain randomization. In contrast, IL methods directly exploit demonstrations: DexDeform [80] extracts latent skills from human demonstrations and fine-tunes them with limited robot data, DefGoalNet [143] predicts goal configurations from few-shot demonstrations conditioned on state and context, and MPD [144] generates movement primitives with diffusion models. To combine both paradigms, DMfD [81] incorporates expert data into RL through an advantage-weighted BC loss, expert-initialized replay buffers, and reset-to-state initialization, achieving stable and efficient policy optimization.

Beyond learning paradigms, DOM has explored diverse strategies spanning geometric modeling, affordance prediction, and structure-aware perception. DeformGS [145] represents deformable objects with canonical Gaussians and learns a time-conditioned deformation field for temporally consistent 3D pose tracking. Affordance-based approaches include DeformerNet [79], which predicts end-effector displacements from partial point clouds, Foresightful Affordance [146], which integrates dense per-pixel affordances with long-horizon value estimation, and APS-Net [137], which ranks standardized folding and flattening trajectories guided by affordance heatmaps. Language-conditioned affordance prediction has also been explored [147]. Finally, estimating object-specific physical properties has emerged as a complementary direction [148], [149]. GenDOM [149] learns a parameter-conditioned policy and leverages a single human demonstration with differentiable simulation to infer unseen object properties, enabling generalization to novel deformable instances.

Challenges. Deformable objects lack a fixed pose, and key deformation regions are often occluded during manipulation. To improve visibility and perception, recent work jointly optimizes camera and manipulator motion, guided by structure-of-interest cues [136]. Deformation dynamics also exhibit delayed responses, complicating real-time control; this is addressed by encoding states into latent spaces and predicting their dynamics. For example, DeformNet [150] encodes object geometry with PointNet and a conditional NeRF, and models temporal evolution with a recurrent state-space model for latent-level MPC. An even greater challenge is modeling topological changes. DoughNet [151] extends latent-space modeling to jointly capture geometric and topological variations from point clouds, enabling long-horizon planning with CEM for tool selection and manipulation.

5) *Mobile Manipulation:* Mobile manipulation is a robotic paradigm that combines navigation and manipulation capabilities within a single system. It allows robots to physically interact with objects beyond a fixed workspace by actively navigating the environment. This integration poses significant challenges in perception, planning, and control, as it requires coordinating whole-body motion, handling long-horizon tasks, and dealing with dynamic, partially observable scenes. Several studies have focused on designing robots specifically for mobile manipulation [152], [155], [156]. As illustrated in



Fig. 5. Tasks in mobile manipulation, figure adapted from [84], [152]–[154].

Figure 5, mobile manipulation is critical for real-world applications such as household assistance, warehouse automation, and service robotics [157].

Non-Learning-based Methods. Traditionally, mobile manipulation is decomposed into two separate problems: navigation and manipulation. Navigation is typically handled by classical path planners, such as grid-based methods (e.g., Dijkstra) and sampling-based planners (e.g., RRT). Manipulation is often addressed using grasp models, motion primitives, or trajectory planning based on point clouds. Some control methods perform joint optimization of navigation and manipulation. For example, Berenson et al. [158] optimize full-body configurations and grasp poses simultaneously, then plan motions via sampling. Chitta et al. [159] coordinate base-arm planning with ROS-based navigation and point cloud grasping. Others embed manipulation goals directly into MPC cost functions [160].

Learning-based Methods. These methods have emerged to learn policies for whole-body control. For RL, Wang et al. [161] use RGB-D inputs to estimate object pose and train a policy that jointly predicts base velocity, arm trajectories, and gripper actions. HarmonicMM [162] extends this by integrating visual and pose information into a unified RL framework, while Wu et al. [82] propose spatial Q-value learning at the map level for navigation guidance. To improve reward signals, Honerkamp et al. [163] design dense rewards based on end-effector reachability, and Causal MoMa [83] introduces causality-aware modeling of control–reward relations to stabilize policy optimization. For IL, MOMA-Force [84] learns motion and force policies from visual-force demonstrations, while HoMeR [154] decomposes tasks into global keypose prediction and local refinement to map end-effector targets to whole-body actions. Wang et al. [164] leverage SAM2-based perception for object-centric IL, and Skill Transformer [85] treats manipulation as a skill prediction problem, learning both skill categories and corresponding low-level actions. Hybrid methods combine IL with RL to enhance generalization. For example, Xiong et al. [86] initialize policies via behavior cloning and refine them through online RL to handle unseen articulated objects.



Fig. 6. Tasks in quadrupedal manipulation, figure adapted from [173]–[177].

Beyond learning paradigms, recent work integrates high-level reasoning and multimodal representations. VLA-based methods such as MoManipVLA [87] process multimodal instructions to predict end-effector waypoints while delegating base motion to a trajectory optimizer. LLM-driven frameworks, including MoMa-LLM [165] and SayPlan [166], combine open-vocabulary language, scene graphs, and reasoning for object search and high-level task planning. Complementary efforts in *3D scene modeling* guide manipulation through active perception: ActPerMoMa [167] optimizes viewpoint selection and grasp reachability via incremental TSDF mapping, while TaMMa [168] employs sparse Gaussian localization and depth completion to generate accurate target poses.

Challenges. In addition to perception–action coordination, dynamic obstacles, and long-horizon decision-making, mobile manipulation faces several additional challenges. The first challenge is enabling effective human–robot collaboration. Ciocarlie et al. [169] developed an assistive system that combines user interfaces, shared control, and autonomy to transform a PR2 robot into an in-home assistant. Extending this line of work, Robi Butler [170] introduces a closed-loop household control framework that supports multimodal interaction, where high-level planners such as LLMs and VLMs enable natural language and gesture-based commands for intuitive and remote collaboration. The second challenge is real-time manipulation during navigation, where robots must coordinate mobility and manipulation under environmental constraints. Whole-body motion control frameworks [171], [172] address this problem by enabling reactive planning and execution for on-the-move manipulation.

6) *Quadrupedal Manipulation:* Quadrupedal manipulation is an emerging paradigm that combines the agile mobility of quadruped robots with the ability to physically interact with objects. Unlike traditional manipulators or mobile bases, quadrupeds can traverse complex and unstructured terrains while maintaining dynamic stability, making them particularly suitable for applications such as search and rescue, field robotics, and autonomous exploration. Representative tasks are illustrated in Figure 6. Manipulation can be realized through several embodiments: whole-body loco-manipulation [93], [94], [178]–[181]; leg-as-manipulator designs that repurpose one or more legs for interaction [91], [174], [182]; back-mounted arms [88], [89], [173], [175], [176], [183]–[191]; and grippers integrated into front legs for simultaneous locomotion and manipulation [192]. Each embodiment introduces unique

challenges in whole-body coordination, dynamic control, and perception-driven planning.

Non-Learning-based Methods. Recent advances in quadrupedal manipulation have explored diverse control strategies, ranging from model-based optimization to learning-based approaches. Optimization-based methods provide interpretable and physically grounded control with strong task generalization. For example, Arcari et al. [185] combine MPC with Bayesian multi-task error learning for real-time dynamics adaptation. Wolfslag et al. [183] incorporate SUF stability metrics and contact constraints into a quadratic programming framework for robust, support-leg-aware planning, a concept further extended by RoLoMa [186] to improve trajectory robustness. LocoMan [192] demonstrates a hardware-control co-design approach, equipping quadrupeds with lightweight front-leg manipulators and employing unified WBC to achieve agile locomotion and precise manipulation.

Learning-based Methods. In the realm of RL, recent research has focused on developing structured and informative control representations. Jeon et al. [178] introduce a hierarchical RL framework that encodes interaction experience, robot morphology, and action history into latent representations to facilitate effective policy learning. Fu et al. [184] decouple leg and arm rewards in the policy gradient formulation to enhance coordination between locomotion and manipulation, while Zhi et al. [191] present a unified force-position controller that estimates forces from perception instead of sensors. Hou et al. [175] incorporate explicit arm kinematics and feasibility-based rewards to promote physically plausible behaviors. Two-stage training schemes have also been explored: RoboDuet [190] sequentially trains locomotion and arm policies with reward adaptation for coordination. GAMMA [89] improves grasp precision by conditioning policies on grasp poses, while Wang et al. [176] propose GORM, a metric for grasp reachability under varying base poses, guiding base movements for optimal grasping. Teacher-student frameworks have further advanced base-arm coordination, as in VBC [88] and Jiang et al. [173], where visual or pose-tracking guidance is distilled into student policies. Other works combine hierarchical and hybrid designs, such as HiLma-Res [179], which uses high-level RL to control Bézier parameters and base motion while relying on hybrid CPG-Bézier controllers for leg movements, and Pedipulate [174], which demonstrates end-to-end RL for single-foot manipulation. In the IL domain, Human2LocoMan [90] enables cross-embodiment transfer from XR-driven human demonstrations using a modular Transformer policy, effectively transferring human manipulation skills to quadrupeds. Hybrid IL-RL frameworks further improve efficiency and generalization. He et al. [91] employ BC to train a high-level planner for grasp trajectories, while coordinating locomotion and single-leg manipulation. WildLMa [92] builds an IL-based skill library from VR demonstrations, sequencing and composing tasks under LLM guidance.

VLA-based frameworks have also emerged for high-level semantic reasoning. QUAR-VLA [93] pioneers the application of VLA models to quadrupeds, while GeRM [94] and MoRE [181] integrate mixture-of-experts architectures with

offline RL to learn generalist visuomotor policies and Q-functions with enhanced generalization and decision quality. QUART-Online [193] advances this line by introducing action-chunk discretization and semantic alignment training, enabling latency-free inference for quadrupedal VLA tasks.

Finally, advances in 3D semantic perception and high-level planning extend quadrupedal capabilities. GeFF [188] performs real-time NeRF-based reconstruction with semantic relevance fields to guide locomotion, while manipulation is executed via learned grasp models. At the task level, LLMs have been combined with policy libraries: Ouyang et al. [180] propose a hierarchical framework where LLMs parse long-horizon, multi-skill instructions into structured subgoals executed by RL-based skills, bridging symbolic reasoning and continuous control.

Challenges. In addition to common challenges such as balance-manipulation coupling, terrain adaptability, and perception delays, real-world deployment remains difficult due to the sim-to-real gap caused by modeling inaccuracies and sensor noise. To mitigate this, fully autonomous RL pipelines have been developed. Mendonca et al. [189] propose a framework that integrates on-policy data collection with continuous training, while ASC [187] addresses long-horizon tasks by decomposing them into modular skills trained via RL, coordinating them through a skill-switching policy jointly trained with IL and RL, and introducing a corrective policy to improve robustness during deployment. Long-horizon task execution itself poses another significant challenge. Cheng et al. [182] propose a stage-wise RL framework that independently learns locomotion and single-leg manipulation skills, which are later composed via a behavior tree to accomplish temporally extended tasks.

7) *Humanoid Manipulation:* Humanoid manipulation involves robotic platforms with human-like morphology, typically including a torso, two arms, and either simplified 2-DoF or fully dexterous hands, with the lower body implemented as either a mobile base or bipedal legs. These systems are designed to perform object interaction tasks in human environments. These systems seek to replicate or extend human manipulation capabilities, enabling actions such as grasping, lifting, tool use, and coordinated bimanual operations, as illustrated in Figure 7. While the humanoid form offers natural compatibility with tools and environments built for humans, it also introduces challenges in balance control, whole-body coordination, and fine motor skills, making humanoid manipulation a central research focus in robotics and embodied intelligence.

Non-Learning-based Methods. Early humanoid control primarily relied on traditional approaches such as rule-based and analytical controllers [198], [199]. Some works also explored the integration of learning-based models into control frameworks. For example, OKAMI [200] leverages 3D human pose estimation from a single human video to infer joint positions and derive executable actions for humanoid robots via inverse kinematics.

Learning-based Methods. The recent trend in humanoid manipulation has shifted from hand-tuned controllers toward fully learning-based approaches. As highlighted in [201], the

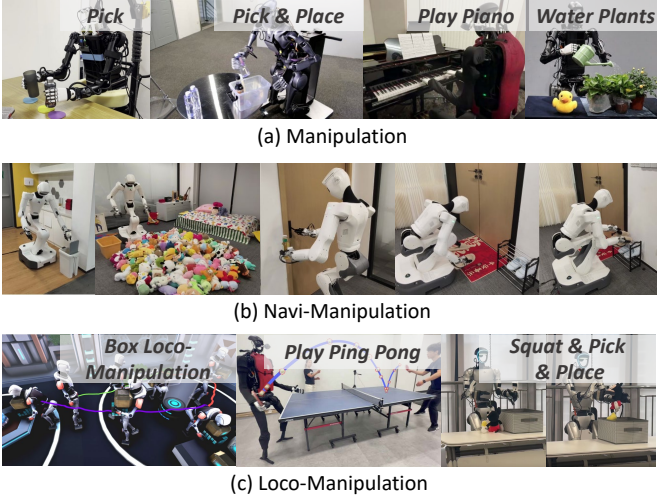


Fig. 7. Tasks in humanoid manipulation, categorized into manipulation [95], [98], [194], navi-manipulation [195], and loco-manipulation [194], [196], [197].

field remains underexplored, with existing methods falling broadly into RL, IL, and VLA frameworks.

For RL-based approaches, FLAM [96] leverages a pre-trained human motion foundation model to score the stability of humanoid poses, using this score as an auxiliary reward to encourage balanced behaviors. Xie et al. [95] combine diffusion-based motion generation with reinforcement learning to achieve whole-body humanoid manipulation. In the domain of IL, TRILL [202] directly maps RGB images to pose commands, while OmniH2O [97] abstracts unified motion goals from diverse modalities (e.g., language, RGB, MoCap, VR) using a diffusion policy. iDP3 [98] adapts the classic DP3 diffusion policy from third-person to egocentric views, enabling teleoperation of the head, torso, and arms. Qiu et al. [196] leverage large-scale egocentric demonstrations to align human and robot actions in a shared representation space, and TACT [203] incorporates tactile feedback to improve manipulation robustness. Hybrid RL+IL strategies have also been proposed. For example, André et al. [99] decouple locomotion and manipulation by training a mid-level trajectory generator with IL to specify motion goals, while using RL to optimize a low-level tracking policy for accurate execution.

Finally, VLA methods aim to develop end-to-end language-conditioned policies for joint locomotion and manipulation. HumanVLA [204] encodes images and language separately, concatenates features, and decodes actions through an MLP. GR00T N1 [100] further integrates visual and tokenized language features via a VLM before feeding them into a diffusion-based head. Humanoid-VLA [101] aligns language and action spaces using cross-attention to fuse visual features, while TrajBooster [197] introduces a loco-manipulation VLA trained with retargeted data from real-world to simulation.

Challenges. In addition to challenges such as multimodal perception, balance–manipulation coupling, and high degrees of freedom, humanoid manipulation also struggles with enabling multi-agent collaboration. For instance, CooHOI [205] exploits object dynamics as an implicit communication channel to achieve coordinated manipulation across multiple agents. To

further address the sim-to-real gap, Lin et al. [206] introduce a generalizable reward formulation and a decoupled RL architecture, which improve sample efficiency through staged training.

C. Generalization

Robotic manipulation generalization can be broadly categorized into three dimensions: **environment generalization**, **task generalization**, and **cross-embodiment generalization**. Environment generalization concerns robustness to variations in conditions such as Sim2Real transfer, spatial transformations, lighting, and background or distractor changes. Task generalization emphasizes maintaining performance across different task configurations, including long-horizon tasks, few-shot or meta-learning, continual learning, and skill composition. Cross-embodiment generalization focuses on transferring skills across robots with diverse morphologies, kinematics, dynamics, or sensing modalities, which is crucial for building general-purpose embodied agents. In what follows, we structure our survey along these three perspectives.

1) *Environment Generalization*: Environment generalization in robotic manipulation refers to the ability of policies trained under specific conditions to maintain performance across varying environments, scenes, viewpoints, and physical settings. Such generalization is challenged by discrepancies between simulation and reality, spatial transformations of objects and cameras, and visual appearance changes such as lighting, backgrounds, and distractors. Existing approaches address these challenges from several complementary perspectives, including sim-to-real and real-to-sim-to-real transfer, geometric equivariance and viewpoint robustness, and robustness to other visual environmental variations.

i) Sim2Real and Real2Sim2Real Generalization

Simulation offers a scalable alternative for training and evaluating manipulation policies, alleviating the cost, operational risk, and safety concerns associated with large-scale real-world data collection. However, discrepancies in dynamics, sensing, visual appearance, and environmental complexity often result in poor transferability, commonly referred to as the sim-to-real gap [215]. Existing approaches can be broadly organized into three paradigms according to how simulated and real-world experience are combined: simulation-only training, sim–real adaptation and co-training, and real2sim2real reconstruction.

Simulation-Only Training. This paradigm assumes that sufficient variability introduced during simulation training can enable policies to transfer directly to the real world without further adaptation. Domain randomization is a representative strategy, where physical parameters such as mass, friction, and damping [216], as well as perceptual factors such as textures [217], lighting [218], and object poses [219], are systematically perturbed to prevent policies from overfitting to a specific simulator configuration. Peng et al. [220] demonstrate that randomized dynamics can substantially improve real-world transfer, while more recent systems such as DexScale [221] scale simulation diversity and augmentation to increasingly complex manipulation settings. Nevertheless, zero-shot transfer remains difficult when simulated variability fails

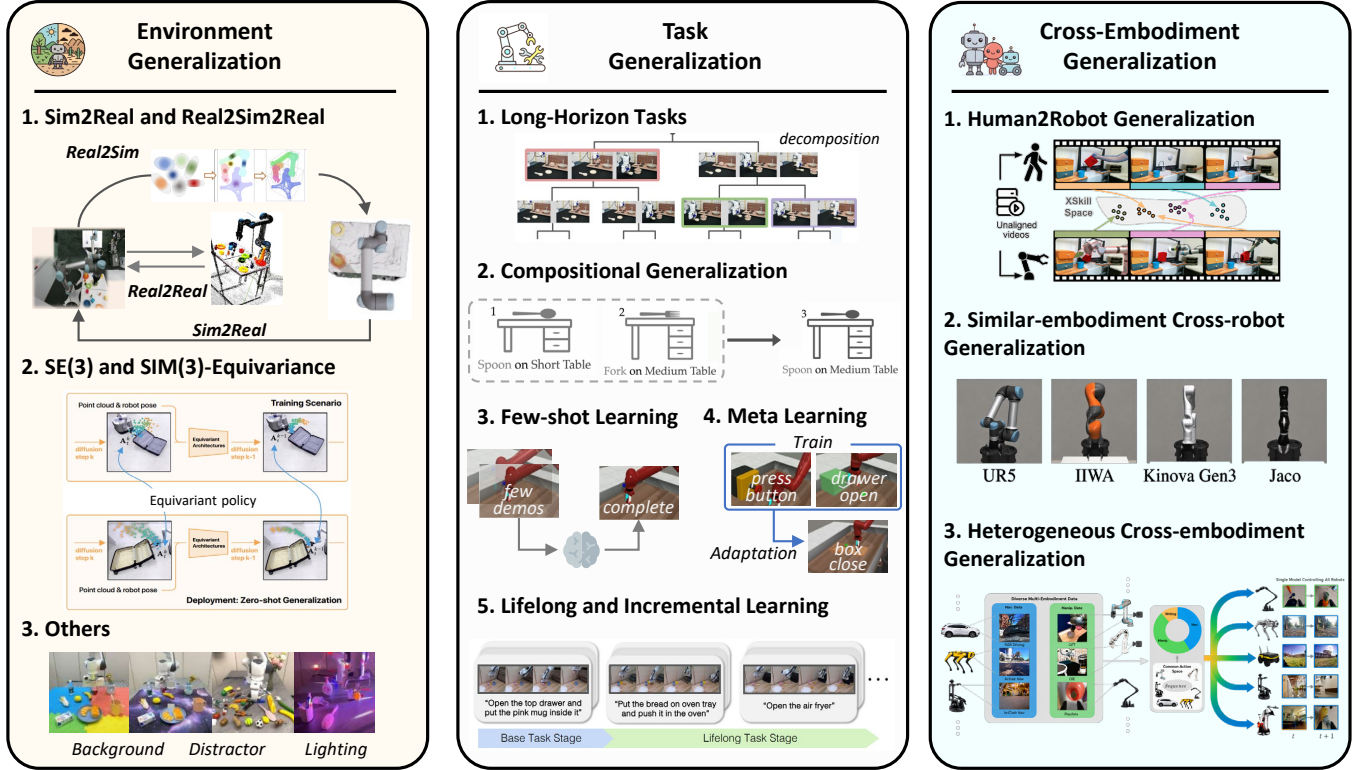


Fig. 8. Overview of generalization in robotic manipulation across three dimensions: environment generalization [207]–[209], task generalization [210], [211], and cross-embodiment generalization [212]–[214]. Representative settings include sim-to-real transfer and robustness to environmental variation, long-horizon and compositional task generalization, and transfer across humans, robots, and embodiments.

to capture fine-grained visual, contact, or dynamic properties of real environments.

Sim-Real Adaptation and Co-Training. Rather than relying exclusively on simulation, another line of work introduces a limited amount of real-world experience to anchor policy learning. Earlier approaches primarily adapt simulation-trained policies through real-world fine-tuning or continual learning; for example, Josifovski et al. [222] study safe continual adaptation under deployment constraints. Natural Language Can Help Bridge the Sim2Real Gap [223] instead uses language descriptions as a shared semantic signal across simulated and real observations, encouraging domain-invariant representations while training with abundant simulated and sparse real demonstrations. More recently, Sim-and-Real Co-Training [224] systematically demonstrates that directly training a shared policy on mixed simulated and real datasets can substantially improve real-world manipulation performance even when noticeable visual and behavioral discrepancies remain between domains. Generalizable Domain Adaptation [225] further aligns the joint distributions of observations and actions across simulation and reality through optimal-transport-based objectives, learning task-relevant domain-invariant representations and enabling simulated experience to improve real-world performance even in scenarios not represented by the limited real demonstrations. These developments shift sim-to-real learning from post-hoc policy correction toward joint exploitation of large-scale simulation and sparse real-world supervision.

Real2Sim2Real Reconstruction. A complementary paradigm

closes the loop by reconstructing simulatable or renderable environments from real-world observations and subsequently using these reconstructions for policy learning [226], [227]. RialTo [226] constructs task-specific digital twins from limited real-world sensing and introduces inverse distillation to transfer real demonstrations into simulation, where reinforcement learning can safely improve policy robustness before deployment back in the real world. Real2Render2Real [227] reduces dependence on both dynamics simulation and physical robot collection by reconstructing object geometry and appearance from smartphone scans and tracking object motion from a single human demonstration, enabling large-scale rendering of robot-compatible training trajectories. Real2Edit2Real [228] further introduces a metric 3D control interface that edits reconstructed scenes and manipulation trajectories and uses depth-conditioned video generation to synthesize geometrically consistent demonstrations under new spatial configurations. These approaches illustrate an emerging transition from manually constructed simulation environments toward real-world-grounded digital reconstruction and editable synthetic environments for scalable policy generalization.

ii) SE(3) and SIM(3)-Equivariance Generalization

A complementary strategy for environment generalization is to explicitly encode geometric symmetry into policy architectures. SE(3)- and SIM(3)-equivariant models constrain representations or actions to transform predictably under changes in translation, rotation, and, for SIM(3), scale, thereby reducing the need to independently observe every spatial configura-

tion during training. EquiBot [208] introduces a SIM(3)-equivariant diffusion policy that improves data efficiency and generalization across changes in object pose and scale. ET-SEED [229] develops an efficient trajectory-level SE(3)-equivariant diffusion policy and relaxes the requirement that every transition in the diffusion process must independently employ expensive equivariant computation. More recently, E3Flow [230] combines spherical-harmonic-based equivariant representations with rectified flow and multimodal visual features, extending geometric equivariance toward more efficient flow-based visuomotor policy learning. Together, these methods demonstrate how explicit geometric structure can provide strong inductive biases for spatial generalization.

Viewpoint Robustness Beyond Explicit Equivariance. Related approaches address viewpoint changes without requiring the policy to satisfy explicit SE(3) or SIM(3) equivariance. Adapt3R [231] combines pretrained semantic visual features with end-effector-relative 3D localization, enabling imitation policies to transfer across unseen camera viewpoints and robot embodiments. Another direction actively changes the observation process itself rather than passively requiring the policy to tolerate arbitrary viewpoints. Vision in Action [232] learns task-dependent searching, tracking, and focusing behaviors from human demonstrations using an actuated robotic neck, allowing the robot to recover informative viewpoints when objects become occluded. SaPaVe [233] extends active perception to VLA-based manipulation by decoupling camera motion from manipulation actions and coordinating semantic camera control with geometry-aware execution. These methods complement explicit equivariance by either constructing viewpoint-robust spatial representations or actively controlling perception to maintain informative observations during manipulation.

iii) Others

Lighting, background, and distractor variations constitute another major source of environment shift, as visuomotor policies can overfit to visual cues that correlate with demonstrations but are irrelevant to task execution. Existing approaches address these variations through augmentation, invariant representation learning, and training across heterogeneous visual contexts. Physically based augmentation methods explicitly vary illumination during training to improve robustness under lighting changes [234]. Decomposing the Generalization Gap [235] systematically analyzes how individual visual factors contribute to imitation-learning generalization failures, highlighting the importance of separating task-relevant information from nuisance variations. Maniwhere [236] combines multi-view representation learning with curriculum-based randomization and augmentation to improve robustness across combinations of visual disturbances and support sim-to-real transfer. Other approaches reduce dependence on incidental scene context through position-invariant regularization [237] or by learning from demonstrations collected under changing contexts [238], [239]. At a larger scale, Diffusion-VLA [240] demonstrates that large multimodal foundation policies can retain manipulation performance under unseen objects, distractors, and new backgrounds, suggesting that broad pretraining can complement explicit augmentation and invariance mechanisms. Collectively, these methods aim to prevent policies

from relying on spurious visual correlations and preserve task-relevant behavior under substantial changes in environmental appearance.

2) *Task Generalization:* Task generalization in robotic manipulation refers to the ability of learned policies to adapt to new, complex, or unseen tasks without extensive retraining. It encompasses the generalization of learned skills, structures, and adaptation mechanisms across variations in task composition, semantics, and temporal scope. Robust task generalization requires policies not only to execute individual skills but also to compose, transfer, and refine them under new configurations and objectives.

i) Generalization for Long-horizon Tasks

Long-horizon robotic manipulation represents a core challenge for embodied intelligence. Unlike short-horizon tasks that typically involve single-skill execution, long-horizon problems demand the coordination of multiple sub-skills, hierarchical reasoning, and temporally abstract decision-making. Robust generalization requires agents not only to plan and execute extended action sequences but also to adapt across variations in task structure, object arrangement, and dynamic environments. Recent advances approach this challenge along three interconnected axes: skill compositionality, semantic task decomposition, and structure-aware representation learning.

Skill Compositionality. At the foundation lies the ability to compose and reuse modular skills [241]–[247]. STAP [245] addresses sequencing by optimizing the feasibility of action chains. Generative frameworks such as BOSS [246] and BLADE [247] leverage large language models to autonomously expand skill libraries and incorporate semantic grounding into action spaces, thereby enabling flexible skill reuse and extension.

Semantic Task Decomposition. Beyond static skill composition, multimodal semantics have been introduced to parse and decompose tasks into modular components [248]–[253]. PALO [249] employs vision-language models to translate high-level task descriptions into reusable sub-tasks, enabling rapid adaptation with minimal supervision. ManipGen [250] integrates local policies that encode invariances in pose, skill order, and scene layout, combining them with foundational models in vision and motion planning to generalize across unseen long-horizon sequences.

Structure-aware Representation Learning. A finer level of generalization is achieved through task-level abstraction and representation learning [254]–[259]. TBBF [257] decomposes complex tasks into primitive therbligs, facilitating efficient action-object mappings and trajectory synthesis. RoboHorizon [258] formalizes a Recognize-Sense-Plan-Act pipeline by fusing LLMs with multi-view world models, addressing sparse reward supervision and perceptual complexity. HD-Space [259] identifies atomic sub-task boundaries from demonstrations, improving sample efficiency and policy robustness with limited data.

ii) Compositional Generalization

Compositional generalization in robotic manipulation emphasizes the ability to solve novel tasks by systematically recombining known skills, objects, and instructions. Recent research explores diverse strategies to achieve this capability.

One line of work focuses on data-driven efficiency, where targeted data collection strategies are designed to maximize coverage of compositional variations with minimal demonstrations, enabling scalable policy training [210]. Another approach grounds policies in programmatic or symbolic structures, allowing robots to leverage modular task representations for systematic recomposition and generalization to unseen task combinations [260]. Complementary efforts investigate policy architectures that explicitly factorize control into entities or modules, facilitating the recombination of learned components to improve adaptability in complex environments [261]. Together, these methods illustrate a growing effort to move beyond rote memorization of tasks toward systematic generalization, enabling robots to flexibly adapt to combinatorial task spaces.

iii) Few-shot Learning

Few-shot learning enables robots to acquire skills from only a handful of demonstrations [262]–[266], alleviating the high cost of large-scale data collection. Research in this area can be broadly grouped into two directions: embedding structured priors and leveraging semantic transfer. One line of work embeds structured inductive biases into perception and control to maximize the utility of limited demonstrations. Domain-invariant constraints, including spatial equivariance, 3D geometry, or action continuity, are incorporated to reduce data requirements and improve generalization. For example, Ren et al. [267] combine point cloud representations with a diffusion-based policy, achieving robust generalization from as few as ten demonstrations. A complementary line of research emphasizes semantic transfer and compositional generalization. These methods extract transferable knowledge from alternative sources, such as human demonstrations or previously solved tasks, and adapt it to new objectives. For instance, YOTO [268] maps human hand motions from video to dual-arm robotic skills, while TOPIC [269] constructs task prompts and a dynamic relation graph to systematically reuse prior experience for new policy learning.

iv) Meta Learning

While closely related to few-shot learning, meta learning distinguishes itself by focusing on the ability to *amortize* task adaptation. Instead of relying directly on structural priors or semantic transfer to generalize from a handful of demonstrations, meta learning trains over a distribution of tasks so that the model acquires a generic adaptation procedure. This paradigm equips robots with rapid adaptability to unseen tasks, even under extremely limited supervision [270]–[276].

Task-Conditioned Meta-Representations. One direction focuses on learning task-conditioned embeddings that serve as meta-representations. Early approaches such as Duan et al. [270] and TecNets [271] encode demonstrations into low-dimensional vectors for policy conditioning. Later works extend this idea with relational graph networks [272] or invariance-based objectives [273], enforcing structural consistency across tasks. These embedding-based approaches highlight the importance of compact task representations in accelerating policy adaptation.

Cross-Modal and Instruction-Driven Adaptation. Another direction emphasizes the integration of additional

modalities and task instructions into the adaptation process. MILLION [274] incorporates natural language into the meta-learning loop, enabling semantically guided adaptation. FISH [275] demonstrates versatile skill acquisition from just one minute of demonstrations, while interaction-warping [276] aligns novel trajectories with past interaction structures to facilitate transfer. Collectively, these works establish meta learning as a principled framework for scalable and data-efficient generalization.

v) Lifelong, Continual and Incremental Learning

Endowing robots with the ability to continuously acquire new capabilities is a key step toward adaptive and autonomous embodied agents. Unlike conventional task generalization, which typically evaluates transfer to a fixed set of unseen tasks, lifelong and incremental learning consider a non-stationary setting in which new tasks or skills arrive sequentially. The central challenge is therefore twofold: newly acquired knowledge should benefit future learning, while previously learned capabilities should be retained without repeated full retraining. In robotic manipulation, existing approaches can be broadly distinguished into lifelong or continual learning, which emphasizes knowledge retention and transfer across a sequence of tasks, and skill-incremental learning, which focuses more explicitly on expanding and updating an evolving repertoire of manipulation skills.

Lifelong and Continual Learning. Lifelong and continual learning aims to retain and reuse knowledge as robots encounter a sequence of new manipulation tasks [277]–[280]. A major direction is to organize manipulation knowledge into reusable modular components. LOTUS [211] continually discovers recurring skills from incoming task demonstrations, incrementally expands or updates a skill library, and composes these skills through a hierarchical meta-controller, enabling forward and backward transfer across sequential tasks. PPL [277] instead represents shared motion primitives as reusable and extensible prompts; pretrained primitive prompts are preserved while new prompts are introduced for incoming skills, allowing previous knowledge to facilitate subsequent learning without overwriting existing capabilities. Complementary approaches focus more directly on preserving learned representations. M2Distill [278] employs multimodal knowledge distillation to maintain consistent representations across sequential tasks and mitigate catastrophic forgetting, while CRIL [279] uses generative replay to preserve experience from previously learned behaviors. Together, these approaches show that continual robot learning increasingly combines modular skill reuse with explicit mechanisms for knowledge preservation, balancing forward transfer to new tasks against retention of previously acquired capabilities.

Incremental Learning. In contrast, skill-incremental learning focuses more explicitly on settings in which robots progressively acquire or refine atomic manipulation skills while preserving their existing skill repertoire. iManip [281] formalizes this setting with a skill-incremental benchmark built on RL Bench [282] and introduces temporal replay together with an Extendable PerceiverIO, whose expandable action prompts accommodate new action primitives while mitigating catastrophic forgetting of previously learned skills. SIL-C [283]

further reveals that incremental skill acquisition introduces an additional challenge beyond forgetting: as the underlying skill library evolves, previously learned high-level policies may become incompatible with newly added or updated skills. To address this issue, SIL-C introduces a lazy-learning interface that dynamically aligns policy-level subtasks with the evolving skill space according to trajectory-distribution similarity, allowing new skills to be incorporated while maintaining compatibility with existing downstream policies. These approaches highlight two complementary aspects of skill-incremental manipulation: extending the policy representation to accommodate new primitives and maintaining a stable interface between an evolving skill library and previously learned task policies.

3) *Cross-Embodiment Generalization*: Cross-embodiment generalization refers to the ability to transfer manipulation knowledge across agents or robotic platforms with different visual appearances, kinematics, action spaces, morphologies, or sensing configurations. Rather than categorizing existing approaches according to specific algorithmic mechanisms, we organize this direction according to the relationship between the source and target embodiments. Specifically, existing studies can be broadly divided into human-to-robot generalization, cross-robot transfer between functionally similar embodiments, and heterogeneous cross-embodiment generalization across substantially different robot morphologies and action spaces.

i) Human-to-Robot Generalization

Human demonstrations provide a scalable source of manipulation experience, but transferring them to robots requires bridging substantial differences in appearance, sensing, morphology, and action spaces. Early approaches therefore seek embodiment-invariant representations or transferable intermediate signals that preserve task semantics while suppressing embodiment-specific motion details. XSkill [212] discovers shared skill prototypes from unlabeled human and robot manipulation videos and maps them to robot actions through a skill-conditioned diffusion policy. Human2Sim2Robot [284] instead extracts object trajectories and hand configurations from a single human RGB-D demonstration and converts them into embodiment-agnostic rewards and initialization priors for reinforcement learning in simulation. X-Sim [285] similarly avoids direct human-to-robot action mapping by reconstructing photorealistic simulation from human videos and using object motion as a transferable learning signal before distilling the resulting behavior into a real-world policy.

More recent methods explicitly align human and robot representations during policy learning. UniSkill [286] learns embodiment-agnostic skill representations from large-scale unlabeled cross-embodiment videos, allowing skills extracted from human video prompts to guide robot policies trained with robot actions. ImMimic [287] maps retargeted human trajectories to robot trajectories and constructs intermediate domains through trajectory interpolation, reducing visual, morphological, and physical discrepancies during co-training. EgoBridge [288] further formulates human-to-robot transfer as domain adaptation and aligns joint latent feature-action distributions using optimal transport guided by trajectory

similarity. Together, these approaches illustrate a progression from indirect motion and object-centric transfer toward explicit human-robot representation alignment.

ii) Similar-Embodiment Cross-Robot Generalization

A second setting considers transfer between robot platforms that retain similar functional structures, such as serial manipulators equipped with comparable end effectors. Although these robots may differ in appearance, degrees of freedom, kinematics, or control dynamics, substantial policy knowledge can often be reused without learning entirely new behavioral abstractions. Mirage [214] enables zero-shot transfer between different robot arms and grippers by separately correcting perceptual and control discrepancies through cross-painting, state alignment, and dynamics compensation. Wang et al. [289] instead project heterogeneous state and action spaces into a shared latent control space, enabling transfer across manipulators with mismatched kinematics. Modularity through Attention [290] takes a complementary approach by decomposing language-conditioned manipulation policies into reusable functional components and robot-specific modules, so that only embodiment-dependent components need to be adapted. These studies show that when source and target robots remain functionally similar, transfer can often be achieved through selective alignment of perception, control, or modular policy components.

iii) Heterogeneous Cross-Embodiment Generalization

The most challenging setting considers substantially heterogeneous embodiments whose observation spaces, action dimensions, kinematic structures, or functional capabilities may differ significantly. One direction addresses this problem through large-scale heterogeneous co-training and shared policy representations. Pushing the Limits of Cross-Embodiment Learning [213] jointly learns from manipulation and navigation data spanning robotic arms, mobile bases, quadrupeds, and aerial robots, demonstrating positive transfer across markedly different embodiments. CrossFormer [291] further represents heterogeneous observations and actions as variable-length token sequences processed by a shared Transformer, supporting manipulation, navigation, locomotion, and aviation within a unified policy. HPT [292] adopts a modular alternative by combining embodiment-specific input stems and output heads with a shared Transformer trunk, allowing heterogeneous visual and proprioceptive data to contribute to common policy pretraining.

A complementary direction constructs explicit structural, functional, or system-level bridges between different morphologies. AnyBimanual [293] transfers pretrained unimanual skills to bimanual manipulation by composing reusable skill representations and aligning arm-specific visual observations. LEGATO [294] introduces a shared grasping tool that provides a common interaction interface across embodiments, allowing manipulation behaviors to be retargeted through embodiment-specific inverse kinematics. CEI [295] explicitly models functional similarity between different robot arms and end effectors and aligns trajectories in 3D space, including transfer between parallel-jaw grippers and dexterous hands. At a higher abstraction level, RoboOS [296] organizes heterogeneous robot capabilities through standardized interfaces

for perception, control, communication, and skill execution, enabling reusable high-level skills to operate across different robotic platforms. Together, these methods show that heterogeneous cross-embodiment transfer can be supported through shared policies, functional correspondence, common physical interfaces, and system-level abstraction.

D. Quantitative Comparison and Empirical Findings

1) *Cross-Benchmark Patterns in Reported Results:* Table III summarizes the reported performance of representative low-level action models on CALVIN [297], LIBERO [298], and SimplerEnv [299]. High-level planning methods are not included in this quantitative aggregation because they typically generate heterogeneous intermediate outputs, such as subgoals, skill calls, programs, keypoints, or geometric constraints, which must be grounded and executed by downstream policies, controllers, or skill libraries. Their end-to-end success therefore reflects the joint effectiveness of planning and execution rather than the quality of the planner alone, making cross-paper task-success rates difficult to compare without a shared executor or standardized planning-specific metrics. This exclusion applies only to the quantitative table.

The reported results provide a useful overview of the performance levels in the literature, but they should not be interpreted as the outcome of a unified reproduction conducted under a fully matched training and evaluation protocol. Although the compared methods generally use the same nominal benchmark tasks and underlying demonstration sources, their effective pipelines may differ in data conversion, image pre-processing, action representation, normalization, checkpoint selection, and evaluation configuration. We therefore use Table III to identify broad benchmark-level patterns, while relying on controlled empirical studies to analyze the effects of individual VLA design choices.

a) *Different benchmarks measure different capabilities, and model rankings are not stable across them:* CALVIN, LIBERO, and SimplerEnv emphasize different aspects of robotic manipulation. CALVIN evaluates language-conditioned sequential execution, with its principal metric measuring how many consecutive tasks a policy can complete within a long-horizon rollout. LIBERO evaluates task learning and execution across four suites that emphasize spatial relations, object variations, goal specifications, and long-horizon composition, respectively. SimplerEnv instead constructs simulation counterparts of real-robot environments and evaluates policies originally intended for real-world deployment, thereby providing a complementary real-to-sim perspective.

Consequently, performance on one benchmark does not reliably predict performance on another. For example, OpenVLA-OFT reports a higher average success rate than π_0 on LIBERO, whereas π_0 achieves higher reported scores in the Google Robot and WidowX settings of SimplerEnv. Similar ranking changes can be observed for other models and embodiments. These results indicate that current VLA performance is strongly benchmark- and embodiment-dependent. No individual benchmark should therefore be treated as a complete measure of general-purpose manipulation capability.

b) *Several LIBERO suites are approaching saturation, whereas long-horizon execution remains more discriminative:* Recent methods commonly report success rates close to or above 98% on the LIBERO-Spatial, LIBERO-Object, and LIBERO-Goal suites. The differences among advanced models on these suites have consequently become small, reducing their ability to distinguish improvements in modern VLA architectures. In contrast, LIBERO-Long exhibits larger performance gaps and remains more sensitive to temporal consistency, compositional execution, and error accumulation.

This pattern does not imply that spatial, object-level, or goal-conditioned generalization has been fully solved. The reported scores are obtained within a fixed benchmark ecosystem using benchmark-specific demonstrations and evaluation distributions. Rather, the results suggest that short-horizon in-suite task execution is becoming insufficient as the sole evaluation criterion and that harder long-horizon, out-of-distribution, and cross-embodiment settings are increasingly necessary for meaningful model comparison.

c) *Parameter count is not a sufficient explanation of current VLA performance:* The reported results do not exhibit a simple monotonic relationship between model size and task success. For example, the 0.9B-parameter X-VLA reports higher scores than several 3B- and 7B-parameter models on LIBERO and some SimplerEnv settings. This observation suggests that parameter count alone does not determine downstream manipulation performance.

However, it should not be interpreted as evidence that smaller models are intrinsically superior. Model size is confounded with the choice of vision-language backbone, robot pretraining data, policy architecture, action representation, optimization objective, and fine-tuning recipe. Moreover, many current benchmarks place substantial emphasis on adapting to a fixed set of benchmark tasks. Their scores may therefore reflect not only general-purpose representation capacity but also the efficiency with which a model specializes to the benchmark distribution.

2) *Controlled Evidence on VLA Design Choices:* Cross-paper differences in Table III cannot be causally attributed to any individual design choice because the compared models differ simultaneously in their backbones, training data, action spaces, optimization objectives, and evaluation implementations. To obtain reliable insights, we synthesize controlled comparisons in which one or a small number of factors are varied within a shared experimental setup. The tables below report results using the metrics adopted by the corresponding studies; values from different experimental blocks should not be directly compared as absolute performance. Instead, the comparisons are used to identify recurring trends, conditional findings, and unresolved questions in VLA design.

a) *Decoupling action generation and providing sufficient policy capacity are critical to VLA performance:* A pretrained VLM does not automatically provide an effective interface for robot control, because representations optimized for visual-language understanding must still be transformed into temporally and dynamically grounded actions. As shown in Fig. 9(d), decoupling action generation from the VLM through a separate policy head increases the LIBERO success

TABLE III
QUANTITATIVE COMPARISON OF REPRESENTATIVE VLA AND POLICY MODELS ON CALVIN [297], LIBERO [298], SIMPLERENV [299].

Method	Params.	Year	CALVIN	LIBERO					SimplerEnv		
			ABC→D	Spatial	Object	Goal	Long	Avg	Google Robot VM	Google Robot VA	WidowX VM
OpenVLA [300]	7B	CoRL 2024	3.270	84.7	88.4	79.2	53.7	76.5	45.8	52.4	4.2
CogACT [301]	7B	2024.12	3.250	95.2	98.6	95.2	89.6	94.7	82.7	66.2	51.3
OpenVLA-OFT [302]	7B	RSS 2025	4.100	97.6	98.4	97.9	94.5	97.1	63.0	54.3	31.3
π_0 [303]	3B	RSS 2025	3.870	96.8	98.8	95.8	85.2	94.2	79.6	66.1	69.2
$\pi_{0.5}$ [304]	3B	CoRL 2025	4.313	98.8	98.2	98.0	92.4	96.8	72.7	68.4	69.7
GR00T N1.5 [100]	3B	2025.03	4.010	96.2	94.0	96.0	90.0	94.0	52.4	43.7	61.9
X-VLA [305]	0.9B	ICLR 2026	4.430	98.2	98.6	97.8	97.6	98.1	80.4	75.7	95.8

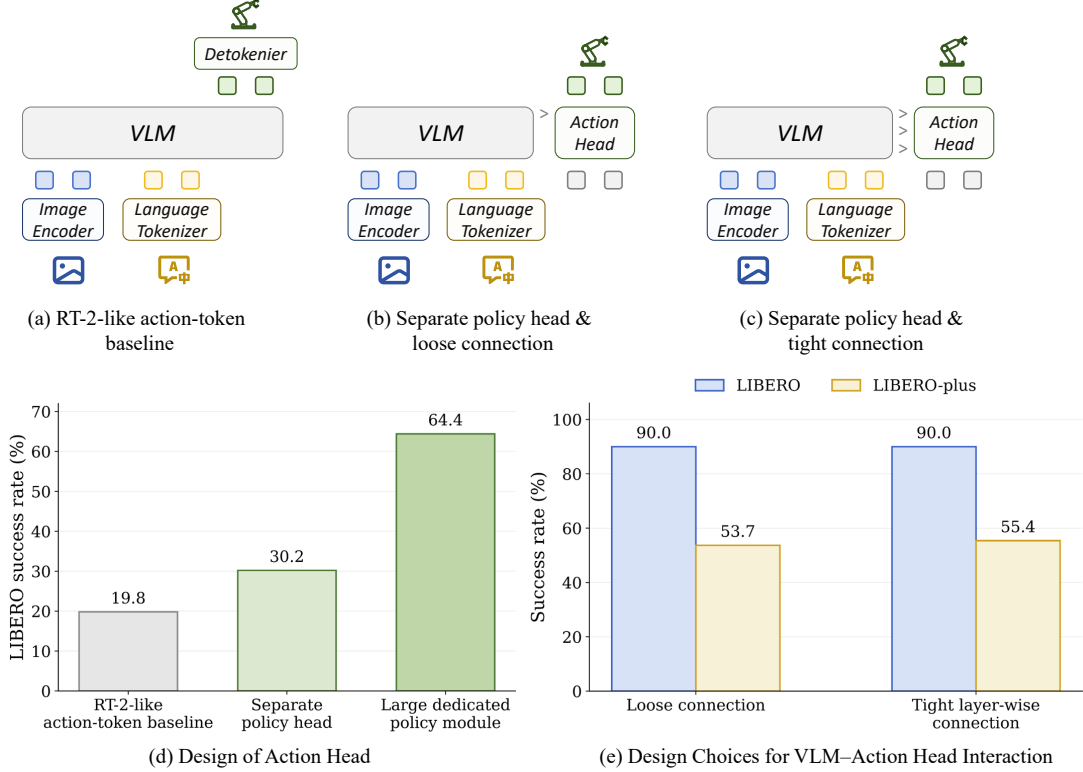


Fig. 9. Controlled evidence on action-module decoupling, policy capacity, and VLM-policy interaction. (a)–(c) illustrate direct action-token prediction, a separate action head with loose coupling, and a separate action head with tight layer-wise coupling, respectively. (d) shows that decoupling action generation from the VLM and increasing policy capacity substantially improve performance. (e) shows that tighter VLM-policy coupling provides only modest, setting-dependent gains once a sufficiently capable policy module is available.

rate from 19.8% to 30.2%, indicating that semantic representation learning and continuous control benefit from specialized computational pathways. Expanding the action head into a larger dedicated policy module further raises performance to 64.4%, demonstrating that action-module capacity constitutes a major bottleneck rather than a secondary implementation detail. Once a sufficiently capable policy module is established, the precise form of VLM-policy interaction has a comparatively smaller effect. As shown in Fig. 9(e), replacing loose coupling with tight layer-wise interaction leaves LIBERO performance unchanged at 90.0%, while producing a modest improvement from 53.7% to 55.4% on LIBERO-plus. This suggests that tighter latent interaction can improve robustness under more challenging visual and environmental variations, but it cannot substitute for adequate action-module capacity. Because the two ablations are conducted at different

stages of model development, their absolute improvements should not be interpreted as directly matched effect sizes. Taken together, the results indicate that architectural separation and sufficient policy capacity are the primary prerequisites for effective vision-language-to-action transfer, whereas tighter VLM-policy coupling provides a smaller, setting-dependent refinement [306].

b) Continuous action formulations generally outperform naive scalar discretization, whereas the relative merits of continuous objectives remain task-dependent: Controlled comparisons indicate that independently quantizing each action dimension into discrete bins can discard fine-grained control information and is particularly vulnerable to error accumulation in long-horizon execution. As summarized in Table IV, replacing per-dimension discrete prediction with continuous actions increases the average number of consecutively com-

TABLE IV

CONTROLLED EVIDENCE ON ACTION REPRESENTATION AND LEARNING OBJECTIVES. HERE, PER-DIMENSION DISCRETE ACTION AND BIN CLASSIFICATION DENOTE THE SAME GENERAL FORMULATION: EACH CONTINUOUS ACTION DIMENSION IS INDEPENDENTLY QUANTIZED INTO SCALAR BINS AND PREDICTED USING A CLASSIFICATION OBJECTIVE, ALTHOUGH THE EXACT BINNING AND MODEL CONFIGURATIONS DIFFER ACROSS THE TWO STUDIES. VQ-VAE CLASSIFICATION INSTEAD REPRESENTS AN ACTION CHUNK USING LEARNED DISCRETE CODEBOOK ENTRIES. DIRECT REGRESSION PREDICTS CONTINUOUS ACTION VALUES WITHOUT DISCRETIZATION, DDIM GENERATES CONTINUOUS ACTION TRAJECTORIES THROUGH ITERATIVE DENOISING, AND FLOW MATCHING LEARNS A CONTINUOUS VECTOR FIELD THAT TRANSPORTS NOISE SAMPLES TOWARD ACTION TRAJECTORIES. IN THE FINAL EXPERIMENTAL BLOCK, MSE+BCE DIRECTLY OPTIMIZES CONTINUOUS MOTION DIMENSIONS WITH MSE AND THE BINARY GRIPPER STATE WITH BCE, WHILE FLOW MATCHING GENERATES THE COMPLETE CONTINUOUS ACTION CHUNK. CALVIN REPORTS AVERAGE CONSECUTIVE TASK LENGTH; LIBERO AND LIBERO-PLUS REPORT SUCCESS RATES (%).

Controlled Factor	Variant	Benchmark	Result
LLaVA action space [307]	Per-dimension discrete action	CALVIN ABCD→D	1.85
	Continuous action	CALVIN ABCD→D	2.37
Action objective [306]	Bin classification	LIBERO / LIBERO-plus	74.6 / 43.4
	VQ-VAE classification	LIBERO / LIBERO-plus	58.8 / 36.5
	Direct regression	LIBERO / LIBERO-plus	85.4 / 48.4
	DDIM	LIBERO / LIBERO-plus	80.0 / 48.3
	Flow matching	LIBERO / LIBERO-plus	80.0 / 45.0
Objective under chunk execution [307]	MSE+BCE	CALVIN ABC→D	3.57
	Flow matching	CALVIN ABC→D	3.68

pleted CALVIN tasks from 1.85 to 2.37. A broader comparison on LIBERO and LIBERO-plus yields a consistent pattern: direct regression, DDIM, and flow matching all outperform the examined classification-based formulations, while VQ-VAE tokenization performs even worse than uniform binning in this setting. The results nevertheless do not establish a universal ordering among continuous objectives. Direct regression obtains the highest LIBERO score and is effectively tied with DDIM on LIBERO-plus, whereas flow matching improves the CALVIN sequence length only marginally over MSE+BCE, from 3.57 to 3.68. The most robust conclusion is therefore that retaining action precision through continuous prediction is more consistently beneficial than adopting any particular regression or generative objective. At the same time, these comparisons concern naive scalar binning and a specific VQ-VAE formulation and should not be interpreted as evidence that discrete action modeling is intrinsically inferior; structured sequence-level tokenizers may preserve temporal dependencies and action precision more effectively than independently quantized action dimensions [306], [307].

c) Action chunking is important, but the action horizon should be neither too short nor unnecessarily long: Jointly predicting multiple future actions is consistently more effective than single-step prediction, indicating that a horizon of one is insufficient to capture the short-term temporal dependencies required for coordinated manipulation. As shown in Figure 10, increasing the horizon from 1 to 4 improves the success rate from 64.4% to 75.4% on LIBERO and from 34.0% to 40.0% on LIBERO-plus. Extending the horizon to 8 retains a clear advantage over single-step prediction, achieving 74.6% and 43.4%, respectively, but does not provide a uniform improvement over horizon 4: it performs slightly worse on standard LIBERO while achieving the best result on LIBERO-plus. These results suggest that action chunks must be sufficiently long to represent locally coherent motion, but that further increasing the horizon yields setting-dependent rather than monotonic gains. An overly short horizon under-models

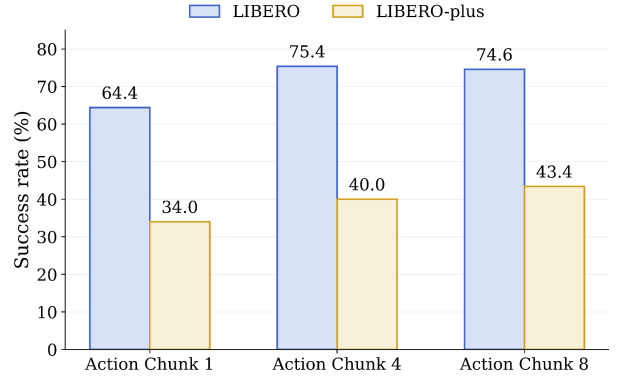


Fig. 10. Effect of action-chunk horizon on policy performance. Action Chunk H denotes jointly predicting a sequence of H future actions. Relative to single-step prediction, both multi-step variants improve performance on LIBERO and LIBERO-plus. However, the preferred horizon depends on the evaluation setting: horizon 4 achieves the highest success rate on standard LIBERO, whereas horizon 8 performs best under the perturbations considered in LIBERO-plus.

temporal structure, whereas an unnecessarily long horizon may reduce the frequency at which the policy can incorporate new observations and correct execution errors, although this mechanism is not directly isolated by the reported ablation. The action horizon should therefore balance temporal coherence with closed-loop responsiveness and be selected according to the task dynamics and evaluation conditions [306].

d) VLM pretraining is essential, but backbone scale alone is insufficient and visual adaptation is critical: Initializing a VLA from a pretrained VLM consistently yields substantially stronger downstream performance than training the same adaptation architecture from random initialization, confirming that pretrained visual-language representations provide an important foundation for manipulation learning. However, as shown in Table V, the resulting control performance does not increase monotonically with backbone size. Within the Qwen3-VL family, the 2B model achieves the strongest

TABLE V

CONTROLLED EVIDENCE ON VLM INITIALIZATION, BACKBONE SCALE, AND MODALITY-LEVEL ADAPTATION. FOR ROWS CONTAINING TWO VALUES, RESULTS ARE REPORTED IN THE ORDER CALVIN / SIMPLERENV. CALVIN REPORTS THE AVERAGE NUMBER OF CONSECUTIVELY COMPLETED TASKS, WHEREAS SIMPLERENV REPORTS SUCCESS RATES (%). RESULTS SHOULD BE COMPARED ONLY WITHIN EACH EXPERIMENTAL BLOCK BECAUSE THE BLOCKS USE DIFFERENT BACKBONES AND ADAPTATION SETTINGS [308].

Controlled Factor	Variant	Evaluation Benchmark	Result
Qwen2.5-VL-3B initialization [308]	Pretrained VLM initialization	CALVIN / SimplerEnv	3.856 / 48.00
	Random initialization	CALVIN / SimplerEnv	1.381 / 15.75
Qwen2.5-VL-7B initialization [308]	Pretrained VLM initialization	CALVIN / SimplerEnv	4.057 / 46.75
	Random initialization	CALVIN / SimplerEnv	1.769 / 18.20
PaliGemma initialization [308]	Pretrained VLM initialization	CALVIN / SimplerEnv	3.506 / 55.25
	Random initialization	CALVIN / SimplerEnv	1.129 / 14.50
Qwen3-VL backbone scale [308]	2B	CALVIN / SimplerEnv	4.142 / 49.0
	4B	CALVIN / SimplerEnv	3.943 / 56.3
	8B	CALVIN / SimplerEnv	4.035 / 58.3
	30B-A3B	CALVIN / SimplerEnv	4.075 / 44.8
Qwen2.5-VL-3B module adaptation [308]	Full fine-tuning	CALVIN / SimplerEnv	3.856 / 48.00
	Freeze vision encoder	CALVIN / SimplerEnv	2.855 / 23.95
	Freeze word embeddings	CALVIN / SimplerEnv	3.849 / 46.88
PaliGemma module adaptation [308]	Full fine-tuning	CALVIN / SimplerEnv	3.506 / 55.25
	Freeze vision encoder	CALVIN / SimplerEnv	0.495 / 13.25
	Freeze word embeddings	CALVIN / SimplerEnv	3.485 / 52.25

CALVIN result of 4.142, whereas the 8B model performs best on SimplerEnv with a success rate of 58.3%; increasing the backbone to 30B-A3B does not improve either benchmark and reduces the SimplerEnv score to 44.8%. Module-level ablations further reveal that this limitation cannot be explained solely by insufficient language-model capacity. Freezing the vision encoder substantially degrades performance, reducing Qwen2.5-VL-3B from 3.856 to 2.855 on CALVIN and from 48.00% to 23.95% on SimplerEnv, while freezing its word embeddings produces only minor changes. The same pattern is more pronounced for PaliGemma, whose performance falls from 3.506 to 0.495 on CALVIN and from 55.25% to 13.25% on SimplerEnv when the vision encoder is frozen. These results indicate that VLM pretraining is necessary but not sufficient: successful VLM-to-VLA transfer depends less on parameter count alone than on adapting pretrained visual representations to the geometric and control-relevant information required by manipulation [308].

e) Robot-data scaling depends on physical alignment and mixture composition rather than data volume alone: Heterogeneous robot pretraining can provide valuable transferable representations, but simply enlarging the pretraining mixture does not guarantee monotonic downstream improvement. As shown in Table VI, under the aligned EEF-relative action representation and frozen-VLM setting, pretraining on OXE increases the LIBERO 5-shot average success rate from 66.9% to 77.3%. However, cumulatively adding real-robot EEF data reduces performance to 73.8%, and further incorporating simulated EEF trajectories decreases it to 72.1%. These results indicate that greater embodiment diversity may introduce interference when differences in data distributions, sensing configurations, and skill frequencies are not sufficiently reconciled. Nevertheless, cross-embodiment pretraining remains useful as an initialization for low-data adaptation, improving

CALVIN few-shot performance by 17.2 percentage points in single-task success. Taken together, the evidence suggests that heterogeneous robot data is most effective when its action semantics are physically aligned and its mixture is matched to the target embodiment and downstream skill distribution, rather than when datasets are pooled solely to maximize nominal scale [307], [309].

f) Unified evaluation largely supports the empirical credibility of published VLA results: The unified harness reduces repository-specific variation through a common model-benchmark communication interface, isolated environments with pinned dependencies, versioned evaluation configurations and simulator assets, fixed random seeds and episode sets, and structured logging of the harness version, benchmark configuration, and per-episode metrics. Under these controlled conditions, the reproduced results remain close to the published values for the models and checkpoints summarized in Table VII. On LIBERO, the absolute discrepancies are generally limited to 0.2–2.1 percentage points, while the reproduced CALVIN results differ by only 0.04–0.13 in average consecutive task length. These small gaps indicate that the published results on LIBERO and CALVIN are broadly reproducible and are unlikely to be driven primarily by independently maintained evaluation scripts. SimplerEnv exhibits larger residual discrepancies of up to 8.0 percentage points. We hypothesize that this larger variation is mainly associated with the greater sensitivity of real-to-sim evaluation to simulator realization and with noisier benchmark conditions, rather than with the evaluated policy alone. Overall, the audit provides positive evidence that published VLA scores are broadly credible at the benchmark level, while also indicating that strict comparisons of small cross-paper differences still require matched evaluation implementations, particularly for real-to-sim benchmarks such as SimplerEnv [310].

TABLE VI

CONTROLLED EVIDENCE ON HETEROGENEOUS ROBOT-DATA SCALING AND CROSS-EMBODIMENT TRANSFER. THE CUMULATIVE-MIXTURE EXPERIMENT EVALUATES LIBERO FEW-SHOT PERFORMANCE UNDER A PHYSICALLY ALIGNED END-EFFECTOR-RELATIVE ACTION REPRESENTATION, WHEREAS THE FINAL EXPERIMENT MEASURES THE EFFECT OF OXE PRETRAINING ON CALVIN FEW-SHOT ADAPTATION. THE RESULTS INDICATE THAT CROSS-EMBODIMENT PRETRAINING CAN IMPROVE LOW-DATA TRANSFER, BUT PROGRESSIVELY ENLARGING A HETEROGENEOUS DATA MIXTURE DOES NOT PRODUCE MONOTONIC GAINS. LIBERO REPORTS AVERAGE SUCCESS RATES (%), WHILE THE CALVIN IMPROVEMENT IS REPORTED AS SINGLE-TASK SUCCESS-RATE POINTS [307], [309].

Controlled Factor	Variant	Evaluation Setting	Result or Change
Cumulative heterogeneous data mixture [309]	Training from scratch	LIBERO few-shot	66.9
	OXE only	LIBERO few-shot	77.3
	OXE + additional real-robot EEf data	LIBERO few-shot	73.8
	OXE + real-robot and simulated EEf data	LIBERO few-shot	72.1
Cross-embodiment initialization [307]	OXE pretraining before target-domain few-shot adaptation	CALVIN few-shot	+17.2

TABLE VII

REPRODUCIBILITY UNDER A UNIFIED VLA EVALUATION HARNESS [310]. EACH REPRODUCED RESULT IS REPORTED TOGETHER WITH ITS DIFFERENCE FROM THE CORRESPONDING PUBLISHED VALUE IN PARENTHESES. BLANK ENTRIES INDICATE SETTINGS THAT WERE NOT EVALUATED IN THE ORIGINAL STUDY; THEREFORE, NO REPRODUCED DIFFERENCE IS REPORTED.

Model	LIBERO	CALVIN	SimplerEnv
OpenVLA [300]	76.2 (−0.3)	—	—
OpenVLA-OFT [302]	96.7 (−0.4)	—	—
GR00T N1.6 [100]	94.9 (−2.1)	—	59.7 (−8.0)
CogACT [301]	94.7 (−0.2)	4.02 (−0.04)	63.5 (−6.0)
X-VLA [305]	97.4 (−0.7)	4.30 (−0.13)	94.8 (−1.0)

3) Supported Conclusions and Remaining Limitations:

a) Supported conclusions: The combined cross-benchmark and controlled evidence supports several comparatively robust conclusions. First, current benchmark scores are capability-specific rather than universally predictive: model rankings vary across CALVIN, LIBERO, and SimplerEnv, while the near-saturation of several standard LIBERO suites increasingly limits their ability to discriminate among advanced policies. Second, policy-side design is a major determinant of VLA performance. Across the examined settings, separating action generation from the VLM, allocating sufficient capacity to the policy module, preserving fine-grained continuous action information, and predicting multi-step action chunks consistently improve upon direct action-token prediction, naive scalar discretization, and single-step control. Third, VLM pretraining provides a strong foundation for downstream policy learning, but neither backbone size nor generic vision-language capability alone reliably predicts manipulation performance; effective transfer additionally requires adapting visual representations to control-relevant geometry and state changes. Fourth, robot-data scaling is governed by action-space alignment, mixture composition, and downstream relevance rather than nominal data volume alone. Finally, unified reproduction recovers most published LIBERO and CALVIN results with relatively small discrepancies, providing positive evidence that the reported benchmark-level performance is broadly credible within the audited models and settings [306]–[310].

b) Setting-dependent findings: Several design choices exhibit conditional rather than universal advantages. Continuous action formulations generally outperform the particular

scalar-binning and VQ-VAE formulations examined here, but the available evidence does not establish a consistent ordering among direct regression, diffusion, and flow matching. Likewise, action chunking is clearly preferable to single-step prediction in the reported ablation, yet the best horizon varies across standard and perturbed evaluation settings. Increasing VLM-policy interaction beyond a sufficiently expressive policy module yields only modest gains, and expanding a heterogeneous robot-data mixture can either improve transfer or introduce interference depending on its physical and statistical alignment with the target setting. Backbone scale, action horizon, generative objective, coupling strategy, and data-mixture size should therefore be treated as jointly dependent design variables rather than independently scalable quantities.

c) Remaining limitations: These conclusions should be interpreted as empirical design guidance rather than universal laws. The controlled evidence is drawn from a small number of studies and remains concentrated in the CALVIN, LIBERO, and SimplerEnv ecosystems. The reported ablations are not based on a common backbone, training budget, or factorial experimental design, and interactions among policy capacity, action representation, temporal horizon, pretraining data, and optimization objective are therefore not fully isolated. Moreover, many studies report point estimates without matched confidence intervals or repeated-seed statistics, limiting the interpretation of small numerical differences. The current synthesis also focuses on low-level VLA policies and does not establish corresponding conclusions for high-level planning, dexterous manipulation, whole-body control, safety-critical interaction, or large-scale real-world deployment. Future empirical studies should combine shared evaluation infrastructure with multi-

seed controlled ablations, harder out-of-distribution and long-horizon tasks, and evaluations spanning both simulation and real robots.

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